

FIN•GUARD COMPLIANCE REPORT

Real-Time Anti-Money Laundering (AML) & Transaction Fraud Graph Analyzer

1. Executive Summary

Modern financial cybercriminals bypass traditional single-card rule triggers by routing funds through multi-account structural networks (mule chains) and breaking down large sums into minor transfers (structuring). FinGuard resolves this capability gap by combining concurrent transaction buffering with an AI ensemble of deep learning neural networks (PyTorch Autoencoders) and statistical partitioning (Isolation Forests), coupled with an in-memory graph analyzer for directed cycle paths.

Forensic Detection Milestone:
During end-to-end pipeline verification, the system successfully isolated a 4-hop circular money laundering loop (ACC_MULE_002 → ACC_MULE_007 → ACC_MULE_003 → ACC_MULE_005 → ACC_MULE_002) in real-time, automatically escalating the risk score to 100% and caching the anomalous profiles in MongoDB.

2. System Architecture & Tech Stack

FinGuard is designed for modularity, scalability, and sub-millisecond screening throughput. The backend is built using FastAPI, running Python's asynchronous tasks, and storing dynamic cache matrices in MongoDB. The frontend compliance dashboard is hosted on Next.js 16 or openable directly via static Vanilla HTML/CSS/JS configurations.

Layer	Component	Functionality
AI/ML Layer	PyTorch Autoencoder & Isolation Forest	Detects spatial statistical vector shifts in transaction behavior.
Graph Engine	In-memory Cycle Path DFS	Traces transaction linkages to detect circular loops and smurfing structures.
Backend API	FastAPI & Asyncio Ingestion Queue	Buffers active transaction streams concurrently and broadcasts WebSocket telemetry.
Database	MongoDB (Motor Async Driver)	Stores transaction logs and caches computed risk velocity matrices.
UI Client	HTML5 Canvas Physics Graph	Draws dynamic force-directed node layouts and highlights anomalies in glowing red vectors.

3. The AI Ingestion Screening Layer

FinGuard evaluates each transaction using a hybrid deep learning model:

- **PyTorch Autoencoder:** Compresses 8 metrics (amount, velocity_1m, velocity_10m, freq_1m, freq_10m, out-degree, in-degree, loop-involvement) into a 4-dimensional latent bottleneck and reconstructs them. Reconstruction Mean Squared Error (MSE) measures vector shift anomaly scores.
- **Isolation Forest:** Outlier partitioning isolates anomalies close to the root of decision trees, calculating statistical outliers.
- **Risk Scale Aggregation:** Computes the final probability as: $\text{Risk} = \max(\text{Autoencoder Risk}, \text{Isolation Forest Risk})$. If a structural loop is detected, the risk score is automatically escalated to 98%.

4. Graph Topology & Cycle Isolation

FinGuard constructs a directed network graph of active transactions occurring within a rolling 10-minute window. Upon ingesting a transfer $S \rightarrow R$, the engine runs a DFS search from R to S. If a path is found, adding the connection closes a money laundering loop. The system flags the loop (e.g. $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$) and isolates the entire path as a glowing red neon vector with flowing cash particles on the UI Canvas.